UNIVERSITI TEKNOLOGI MALAYSIA

MID-TERM EXAMINATION SEMESTER 2 2016/2017

CODE OF SUBJECT	**: SCSJ 2253**
NAME OF SUBJECT	**: REQUIREMENTS ENGINEERING AND SOFTWARE MODELING**
YEAR / COURSE	**: 2 SCSJ**
TIME	**: 9.00 – 11.30 AM (2 HOURS 30 MINUTES)**
DATE	**: 24 MARCH 2016**
VENUE	**: BILIK KULIAH 1 & 2, N28**

INSTRUCTIONS TO THE STUDENTS:

This test book consists of 3 sections:

SECTION A: True/False	10 marks
SECTION B: Structured questions	50 marks
SECTION C: Problem solving	40 marks

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET.

This examination book consists of **10 (SIX) printed pages** including this page.

SECTION A – TRUE/FALSE QUESTIONS [10 marks]

There are **5** questions in this section. For each statement given in this section, identify whether the statement is **TRUE/FALSE.** For the **FALSE** answer, you are required to give a reason for it. Each question carries **2** marks.

1. The following list of statements are considered as constraint requirements
   (i) The system must be available 95% of the time.
   (ii) The system must provide clerk with the ability to save the inventory balance locally.
   (iii) The system shall be able to display the offline help manual to the users.

2. The sources of requirements are found in the system context. The system context does not include the estimated project cost.

3. Goals are a stakeholder's (e.g., a person's or an organization's) description of a characteristic property of the system to be developed or the development project.

4. Brainstorming and change of perspective are two techniques that best suited for eliciting dissatisfiers factor of customer satisfactions.

5. A large, regional railway company with 70 stations is purchasing a new communication device for station employees. Observation of works of selected stakeholders at each station is a suitable elicitation technique to determine the requirements.

SECTION B – STRUCTURED QUESTIONS [50 marks]

Instruction: There are **7** questions. The marks for each part of the question is as indicated.

Question 1 until **4** are based on the following description of the Service Level Reporting System.

Food Service Supplies (FSS) is a Malaysia-based provider of perishable foodstuffs to the catering industry. Their main clients are hotels, schools and hospitals, although they have also recently started supplying professional catering companies.

Orders are received via three channels: 1) calls directly from the customer to a telesales function, 2) online orders placed directly by the customer via the internet and 3) receipt of paper order forms into the sales department's data processing team.

Upon receipt of an order, it is entered into the Sales Order Processing (SOP) system where it is validated and sent to the Stock Management System (SMS) used by the warehouse. The SMS manages the picking process and, when an order has been picked, the details are transferred to the Traffic Management System (TMS). The TMS is used by the Traffic Office to schedule and track the dispatch and delivery process. Finally, the signed Delivery Note (signed by the customer upon receipt of the goods) is returned by the delivery driver to the Accounts Payable team where it is processed in the Central Accounts System (CAS) and an invoice generated for the customer.

As part of a new continuous service improvement initiative FSS Managing Director Alan Jones has initiated a project to investigate the requirements for a proposed Service Level Reporting (SLR) solution to track service level failures and report on key performance indicators across the organization. Examples of service failures are:

- Misinterpretation of customer order
- Damage to goods during distribution
- Incorrect invoice (e.g. charge for products not received)

It has been decided that the scope of the initial solution should be restricted to the following organizational functions: sales, warehousing, distribution and accounts. An organization chart of these areas is provided in **Figure 1**.

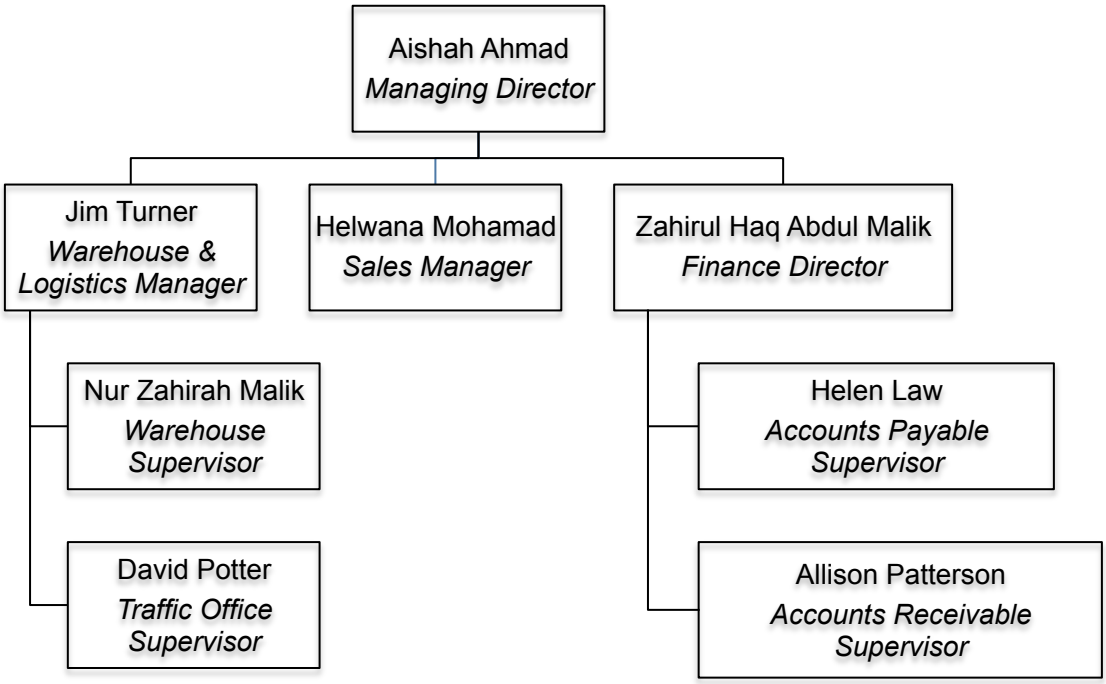


Figure 1: FSS Organization Chart

Question 1: [3 marks]
Based on the new SLR system, explain **TWO (2)** reasons why requirement elicitation activity is difficult.

Question 2: [3 marks]
It has been suggested that a workshop **SHOULD** be used to obtain the high level requirements for the new SLR system. Give **THREE (3)** reasons why a workshop would be a useful technique for achieving this.

Question 3: [4.5 marks]
Identify **THREE (3)** stakeholders that you believe **SHOULD** be included in the workshop and explain why they **SHOULD** attend.

Question 4: [4.5 marks]
Following the workshop, it will be necessary to explore the high level requirements in more detail with each stakeholder individually. For each stakeholder identified in Question 3 suggest a possible elicitation technique that you **COULD** use to define their detailed requirements and explain why the technique would be suitable.

Question 5: [10 marks]

For each requirement, state whether it is Functional **[F]** requirement, Quality **[Q]**, or Constraints **[C]** requirement. **Justify the reason**.

Requirement specification	Answer:
i. The pilot shall be able to view the airspeed from the navigator display	**Type:** **Reason:**
ii. The aircraft shall be able to cruise at a steady airspeed of 800 kilometers per hour at 10,000 meters altitude	**Type:** **Reason:**
iii. All materials used in the passenger compartment shall resist ignition under applied heat to FAA standard 123-45 from US-OSHA board agency.	**Type:** **Reason:**
iv. The Lestari-PSZ system shall allow library patron to check-out the borrow book using the Self-Checkout Machines.	**Type:** **Reason:**
v. The new feature of Daily Alert Trigger in MySelf-Monitoring Apps. must be implemented using HTML5 and jQuery language to support cross-platform development.	**Type:** **Reason:**

Question 6: [12 marks]

The question 6 (i) until (iii) is based on the following description of the Hospital Management System.

The Hospital Management System (HMS)

Hospital Management System (HMS) is a self-contained system that manages activities of the hospital such as bed assignment, operations scheduling, personnel management and administrative issues. The system will be used to allocate beds to patients on a priority basis, and to assign doctors to patients in designated wards as need arises. Doctors will also use the system to keep track of the patients assigned to them. Nurses who are in direct contact with the patients will use the system to keep track of available beds, the patients in the different wards, and the types of medication required for each patient. The functionality of the HMS is illustrated in **Figure 2**. The current system in use is a paper-based system. It is too slow and cannot provide updated lists of patients within a reasonable timeframe. Doctors must make rounds to pick up patients' treatment cards in order to know whether they have cases to treat or not. The intentions of the system are to reduce over-time pay and increase the number of patients that can be treated accurately.

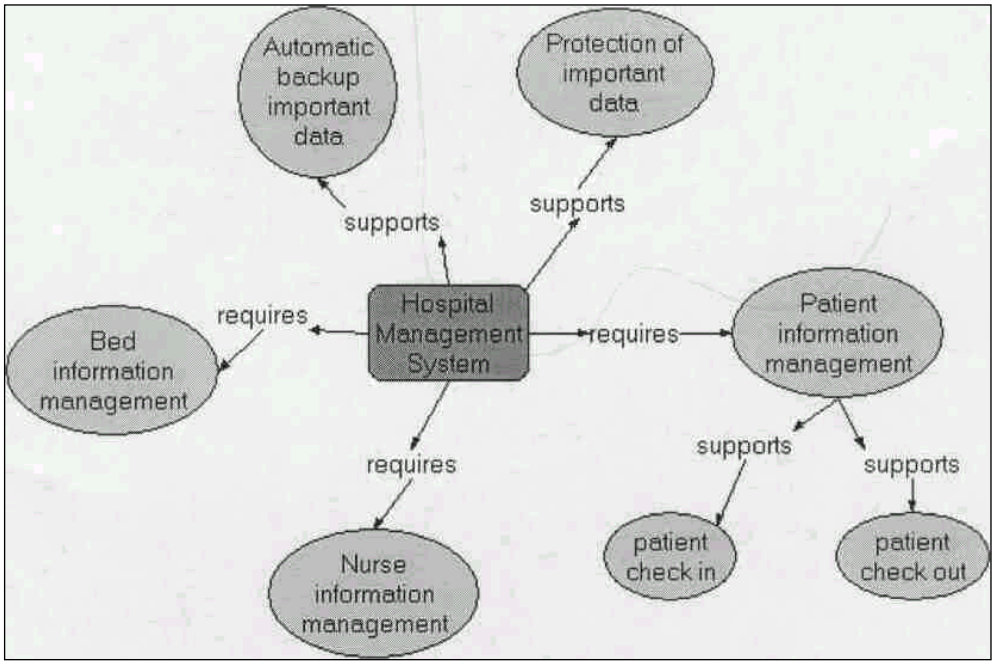


Figure 2: HMS Functionalities

(i) Based on the HMS case study, discuss TWO reasons **WHY** capturing the right and complete requirements is important in developing the HMS system. ***(4 marks)***

(ii) Identify **FIVE (5)** context objects. ***(5 marks)***

(iii) Identify **ONE (1)** grey zone, or, a grey zone object, **AND** provide a **reason** to your answer. ***(3 marks)***

Question 7: [13 marks]

(i) Discuss **TWO (2)** advantages of classifying identified requirements according to Kano Model. ***(3 marks)***

(ii) Based on the following statement, answer the following questions.

> Your company is targeting to produce smart watch software that shall be introduced in the year **2018**. Your company then would like to classify system features of smart watch software using the Kano model.

Categorize all the following features in **Table 1** according to the Kano classification schema: Delighter (Excitement), Performance, and Basic.

(10 marks)

Table 1: Features for smart watch software in year 2018

Features	Kano classification
F1: The smart watch shall be able to function as portable media players, with FM radio and playback of digital audio and video files via a Bluetooth or USB headset	
F2: The smart watch shall have feature full mobile phone capability, and can make or answer phone calls or text messages.	
F3: The smart watch shall have an electronic visual display, either backlit LCD or OLED.	
F4: The smart watch shall be able to support wireless technologies like Bluetooth, Wi-Fi, and GPS.	
F5: The smart watch shall have fitness tracking feature	
F6: The smart watch battery capacity shall be at least one month by high usage.	
F7: The smart watch shall be able to interact and navigate with Virtual Reality menus.	
F8: The smart watch shall have an option to change the watch face every day.	
F9: The smart watch shall be able to turn to standby mode when it is not in use.	
F10: The smart watch shall be waterproof.	

SECTION C – PROBLEM SOLVING [40 marks]

All 4 (FOUR) questions in this section are based on the following case study of dengue monitoring system project proposal.

Your company has won the project tender offered by Ministry of Health (MoH) Malaysia to reinvent the existing dengue monitoring operations by introducing the new application (app) platform using the latest technology of mobile devices. In Malaysia, entomological monitoring is frequently applied in governing the density changes and distribution of the vector in order to effectively response and control of dengue disease outbreaks. The team of Entomology and Pest Unit (EPU) from the Johor Bahru (JB) Health Office is selected by the MoH to be as one of the health office branches as a pilot case study for the newly proposed system.

The proposed dengue-entomological monitoring system is expected to be integrated as an interactive hybrid mobile and web applications that aim to cost-effectively support the EPU team in their fieldwork; especially in capturing and managing surveillance data related to dengue outbreaks. The key features of the application are two fold: (i) the automated reporting suspected dengue cases and (ii) effective data collection and report generation for dengue surveillance and control. The objectives of proposed system are to increase the efficiency of the EPU team by systematically and automatically record and save data collected from suspected dengue case reports and ovitrapping installation operations.

Based on the fieldwork observations during elicitation activity, the EPU team is currently using conventional and manual procedures to perform the dengue monitoring operational tasks. Firstly, the reporters will use phone call, fax or email to lodge report forms of the suspected dengue cases to the JB Health Office. The reporters are basically the verified and registered health practitioners such as government hospitals, private hospitals and the panel clinics. The report form is not up to the standard, which some important and required fields are missing and the data collected are inconsistent.

After the suspected report cases were received, the Johor Bahru Health Office needs to process the reported cases manually and validate them. After being validated by a public health specialist, the validated cases will be notified to EPU team. Entomologists, public health assistant (PHA) and general assistant (GA) are the members in the EPU team.

Next, the entomologist will open a case file and give command to PHA and GA to perform the ovitrapping installation to the locality (suspected case location). *(Note: Ovitrap also known as oviposition trap is a technique use to monitor the Aedes mosquito population and distribution in term of the number of larvae and eggs that was directly collected from the field/locality. Ovitrap is a wide-mouth 0.5 litre glass jar painted black with a piece of hardboard paddle submerged into cup)*.

During the ovitrapping installation, PHA needs to determine the suitable mosquitoes breeding habitats/sites. At least 60 ovitrap cups will be installed approximately within 200 metres radius of the suspected locality. The ovitrap cups will be located at the localities at least 7 days. After 7 days, PHA and GA will collect the cups and calculate the quantity of

positive ovitraps. The calculation process is conducted in the laboratory environment. The percentage of positive ovitraps provides the simplest index of density levels. The related data (i.e. cups ids, houses/premises address, markers, outdoor/indoor locations, name & contact no of peoples, GPS coordinates) for each of the installed ovitraps is recorded in the manual forms during the operation.

When the EPU team complete operation and going back to the JB Health Office, these recorded data in the manual forms were then transferred to the spreadsheet files using Microsoft Excel. The laboratory results for ovitrap analysis such as number of eggs, and number of larvae for each ovitrap cups as well as the index of density are also updated into the spreadsheet files. Transferring this large amount of data from paper-based filed records is a very time consuming, tedious, and error-prone process.

Question 1: [10 Marks]
Produce a **Swimlane diagram** to represent the ***as-is*** manual activities and processes related to dengue monitoring for EPU team of JB health office.

Question 2: [10 Marks]
One of the possible dengue monitoring process that could be automated is the updating ovitrap data during the installation procedure. The automated process must cover from the process of the public health assistant (PHA) enters the related data of cups ids, houses/premises address, markers, outdoor/indoor locations, name & contact no of peoples, GPS coordinates) for each of the installed ovitrap cups at the locality using the mobile devices. Apart from that, the laboratory results for ovitrap analysis such as number of eggs, and number of larvae for each ovitrap cups – are also possible to be updated using the new system.

Produce a **Swimlane diagram** to represent the ***to-be*** automated process for the updating the ovitrap data.

Question 3: [5 Marks]
Produce a **Goal model** to describe the decomposition of the high level visions for the proposed dengue-entomological monitoring system.

Question 4: [5 marks]
Produce a **use case diagram** for the newly proposed dengue-entomological monitoring system.

Question 5: [10 marks]

Produce a complete **use case specification** for the updating the ovitrap data use case. Use the table format with the required details:

Use case ID	
Use case name	
Description	
Actor (s)	
Pre-condition	
Normal Flow	
Alternative/ Exceptions Flow	
Post-condition	

-END-